Jeongseob Ahn

Assistant Professor
Department of Software and Computer Engineering
Ajou University
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Employment

Ajou University, Suwon, South Korea . . . . September 2017–Present

Assistant Professor in Department of Software and Computer Engineering

Research group: Computer Systems Laboratory

Oracle Labs, Belmont, CA . . . . February 2016 –July 2017

Researcher & Engineer

I participated in a research project building a HW accelerated big data analytics (called RAPID)

The University of Michigan, Ann Arbor, MI . . . . February 2015–January 2016

Research Fellow in Computer Science and Engineering

Education

KAIST, Daejeon, South Korea . . . . February 2011–February 2015

Ph.D. in Computer Science

Advisor: Professor Jaehyuk Huh

Dissertation: Architectural Supports for High Performance Address Translation and Processor Scheduling in Virtualized Systems

KAIST, Daejeon, South Korea . . . . February 2009–February 2011

M.S. in Computer Science

Advisor: Professor Jaehyuk Huh

Thesis: Scalable Subspace Snooping: Exploiting Temporal Sharing Stability

Dongguk University, Seoul, South Korea . . . . March 2004–February 2009

B.S. in Computer Science and Engineering

Research Interests

Interested in all computer systems and architecture related problems in general. In particular, my research focus is the interaction of system software and architecture for building high performance AI and ML systems.

Publications

Conferences					Journals	
ISCA	MICRO	EuroSys	PACT	ISLPED	TC	TACO
1	2	2	1	1	3	1

Table 1: Publications in top/major CS conferences and journals

The asterisk (*) denotes the role of the corresponding author after joining Ajou Univ.

Conference Proceedings

[1] **GrandSLAm: Guaranteeing SLAs for Jobs in Microservices Execution Frameworks**.
Ram Srivatsa Kannan, Lavanya Subramanian, Ashwin Raju, **Jeongseob Ahn***, Jason Mars, and Lingjia Tang.
In *Proceedings of the European Conference on Computer Systems (**EuroSys**)*, March 2019.

[2] **A Load Balancing Technique for Memory Channels**.
Byoungchan Oh, Nam Sung Kim, **Jeongseob Ahn**, Bingchao Li, Ronald Dreslinski, and Trevor Mudge.
In *Proceedings of the International Symposium on Memory Systems (**MEMSYS**)*, October 2018.

[3] **Accelerating Critical OS Services in Virtualized Systems with Flexible Micro-sliced Cores**.
Jeongseob Ahn*, Chang Hyun Park, Taekyoung Heo, and Jaehyuk Huh.
In *Proceedings of the European Conference on Computer Systems (**EuroSys**)*, April 2018.

[4] **How Much Computation Power do you need for Near-Data Processing in Cloud?**
Namhyung Kim, **Jeongseob Ahn**, Sungpack Hong, Hassan Chafi, and Kiyoung Choi.
In *Workshop on Architectures and Systems For Big Data (**ASBD**)*, June 2017.

[5] **Enhancing DRAM Self-Refresh Modes for Idle Power Reduction**.
Byoungchan Oh, Nilmini Abeyratne, **Jeongseob Ahn**, Ronald Dreslinski, and Trevor Mudge.
In *Proceedings of the International Symposium on Low Power Electronics and Design (**ISLPED**)*, June 2016.

[6] **Micro-sliced Virtual Processors to Hide the Effect of Discontinuous CPU Availability for Consolidated Systems**.
Jeongseob Ahn, Chang Hyun Park, and Jaehyuk Huh.
In *Proceedings of the International Symposium on Microarchitecture (**MICRO**)*, December 2014.

[7] **Revisiting Hardware-Assisted Page Walks for Virtualized Systems**.
Jeongseob Ahn, Seongwook Jin, and Jaehyuk Huh.
In *Proceedings of the International Symposium on Computer Architecture (**ISCA**)*, June 2012.

[8] **Dynamic Virtual Machine Scheduling in Clouds for Architectural Shared Resources**.
Jeongseob Ahn, Changdae Kim, Jaeung Han, Young-ri Choi, and Jaehyuk Huh.
In *Proceedings of the USENIX Workshop on Hot Topics in Cloud Computing (**HotCloud**)*, June 2012.

[9] **Architectural Support for Secure Virtualization under a Vulnerable Hypervisor**.
Seongwook Jin, **Jeongseob Ahn**, Sanghoon Cha, and Jaehyuk Huh.
In *Proceedings of the International Symposium on Microarchitecture (**MICRO**)*, December 2011.

[10] **Subspace Snooping: Filtering Snoops with Operating System Support**.
Daehoon Kim, **Jeongseob Ahn**, Jaehong Kim, and Jaehyuk Huh.
In *Proceedings of the International Conference on Parallel Architectures and Compilation Techniques (**PACT**)*, September 2010.

[11] **The Effect of Multi-core on HPC Applications in Virtualized Systems**.
Jaeung Han, **Jeongseob Ahn**, Changdae Kim, Youngjin Kwon, Young-ri Choi, and Jaehyuk Huh.
In *Workshop on Virtualization and High-Performance Cloud Computing (**VHPC**)*, August 2010.

Journal Articles

[12] **LightCert: On Designing a Lighter Certificate for Resource-limited Internet of Things Devices**.
Hyuksang Kwon, **Jeongseob Ahn**, and JeongGil Ko.
*Transactions on Emerging Telecommunications Technologies (**ETT**)*, 30(10), September 2019.

[13] **Caliper: Interference Estimator for Multi-tenant Environments Sharing Architectural Resources**.
Ram Srivatsa Kannan, Michael Laurenzano, **Jeongseob Ahn***, Jason Mars, and Lingjia Tang.
*ACM Transactions on Architecture and Code Optimization (**TACO**)*, 16(3), May 2019.

[14] **Deduplicating TLB Entries for Shared Pages**.
Jeongseob Ahn*.
*IEICE Electronics Express (**ELEX**)*, 15(14), July 2018.

[15] **Fast Two-level Address Translation for Virtualized Systems**.
Jeongseob Ahn, Seongwook Jin, and Jaehyuk Huh.
*IEEE Transactions on Computers (**TC**)*, 64(10):3461–3474, December 2015.

[16] **H-SVM: Hardware-assisted Secure Virtual Machines under a Vulnerable Hypervisor**.
Seongwook Jin, **Jeongseob Ahn**, Jinho Seol, Sanghoon Cha, Jaehyuk Huh, and Seungryoul Maeng.
*IEEE Transactions on Computers (**TC**)*, 64(10):2833–2846, October 2015.

[17] **Subspace Snooping: Exploiting Temporal Sharing Stability for Snoop Reduction**.
Jeongseob Ahn, Daehoon Kim, Jaehong Kim, and Jaehyuk Huh.
*IEEE Transactions on Computers (**TC**)*, 61(11):1624–1637, November 2012.

Under Review

[18] **Reconciling Time Slice Conflicts of Virtual Machines with Dual Time Slice for Clouds**.
Taeklim Kim, Chang Hyun Park, Jaehyuk Huh, and **Jeongseob Ahn***.
*IEEE Transactions on Parallel and Distributed Systems (**TPDS**)*, August 2019. Under major revision.

[19] **Research on Tiered Memory Hierarchy (not the submission title)**.
Wonkyo Choe, Jonghyeon Kim, Seongwook Jin, and **Jeongseob Ahn***.
In *submission to a computer architecture conference*, 2019.
* Double blind review.

[20] **Research on Trusted Storage (not the submission title)**.
Chang Hyun Park, Jungi Jeong, **Jeongseob Ahn**, and Jaehyuk Huh.
In *submission to a computer security conference*, 2019.
* Double blind review.

Granted Patents

[21] **System and method of reducing traffic among multi-cores used to meet cache coherence**.
Jaehyuk Huh, **Jeongseob Ahn**, Daehoon Kim, and Jaehong Kim.
Patent No.1012331090000, February 2013.

Teaching Experience

Instructor at Ajou University

Semester Year	Course Number	Course Title	Enrollment	Course Rating
Spring 2020	SCE212	Computer Organization and Architecture (Undergraduate)		/5
Fall 2019	SCE212	Computer Organization and Architecture (Undergraduate)	102	4.59/5
Fall 2019	CSE561	Advanced Computer Architecture (Graduate)*	20	4.50/5
Spring 2019	SCE212	Computer Organization and Architecture (Undergraduate)	116	4.37/5
Fall 2018	SCE212	Computer Organization and Architecture (Undergraduate)	110	4.37/5
Fall 2018	CSE515	Advanced Operating Systems (Graduate, w/ JeongGil Ko)*	17	4.30/5
Spring 2018	SCE213	Operating Systems (Undergraduate)	53	4.35/5
Spring 2018	ICT331	Computer Organization (Undergraduate)	28	3.70/5
Fall 2017	SCE212	Computer Organization and Architecture (Undergraduate)*	62	4.08/5
Fall 2017	CSE561	Advanced Computer Architecture (Graduate)*	14	4.58/5

Table 2: The asterisk (*) denotes the lectures offered by English.

Teaching Assistant at KAIST

CS230 System Programming .... Fall 2014, Fall 2011

CS310 Embedded Computer Systems .... Fall 2012

CS510 Computer Architecture .... Spring 2012, Spring 2010

CS311 Computer Organization .... Sping 2011, Fall 2009

Research Grants

Morphable Memory Hierarchy for Supporting Diverse Heterogeneous Memory in Clouds

Agency/Company: National Research Foundation of Korea (NRF)
Total Amount: 340,000,000 KRW
Role: PI
Period: March 2019 - February 2022
Share: 100%

Exploring Convergence Techniques for Future Cloud Computing

Agency/Company: Ajou University
Total Amount: 30,000,000 KRW
Role: PI
Period: October 2017 - September 2019
Share: 100%

Resource Management Techniques for Cloud Computing

Agency/Company: National Research Foundation of Korea (NRF)
Total Amount: 60,000,000 KRW
Role: PI
Period: September 2017 - August 2019
Share: 100%

Research on Highly Efficient Data Analytics Platform for Large-Scale Machine Learning

Agency/Company: National Research Foundation of Korea (NRF)
Total Amount: 13,550,000 KRW
Role: PI

Period: November 2017 - February 2018
Share: 100%

Honors and Awards

Teaching Excellence Award at Ajou University 2020

Outstanding Dissertation Award in Computer Science at KAIST 2015

National Full Scholarship for Graduate Study .... February 2009 - February 2015

Outstanding Teaching Assistant Award at KAIST .... Fall, 2012

Academic Scholarship at Dongguk University .... Fall 2017, Spring 2018

Professional Activities

Conference External Reviewer

ACM/IEEE International Symposium on Computer Architecture (ISCA) 2019

IEEE International Symposium on High-Performance Computer Architecture (HPCA) 2016, 2013

ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) 2016

IEEE/ACM International Symposium on Microarchitecture (MICRO) 2015

Journal Reviewer

ACM Transactions on Computer Systems (TOCS) 2019

ACM Transactions on Embedded Computing Systems (TECS) 2019

IEEE Micro 2019, 2020

ACM Transactions on Design Automation of Electronic Systems (TODAES) 2016

IEEE Transactions on Cloud Computing (TCC) 2015

IEEE Transactions on Parallel and Distributed Systems (TPDS) 2012

Invited Talks

Cross-layer System Design for Cloud Computing

Korea University, Seoul, South Korea .... Nov 2019

Challenges and Opportunities in the Era of Heterogeneous Memory Systems

ETRI, Daejeon, South Korea .... Dec 2018

Optimizing Consolidated Systems with Flexible Micro-sliced Cores

UNIST, Ulsan, South Korea .... Sep 2018

ETRI, Daejeon, South Korea .... May 2018

Seoul National University, Seoul, South Korea .... May 2018

Flexible Micro-sliced Cores for Virtualized Systems

KIISE Computer System Society, Weonju, South Korea .... January 2018

Architectural Support for High Performance Cloud Computing

Ajou University, Suwon, South Korea .... May 2017

Architectural Support for High Performance Virtualized Systems

ARM Research, Austin, TX .... November 2015

IBM Austin Research Lab, Austin, TX .... November 2015

VMware Research Group, Palo Alto, CA .... October 2015

System Virtualization and Micro-sliced Virtual Processors

The University of Michigan, Ann Arbor, MI .... March 2015

Student Supervision

Graduate Students

Jungmo Ahn (Ph.D. program) co-advised with Prof. JeongGil Ko

Jinwoo Jeong (M.S. program)

Jonghyeon Kim (M.S. program)

Wonkyo Choe (M.S. program)

Undergraduate Students

Taeklim Kim (Winter 2017 - current)

Alumni

Jinwoo Hwang (B.S. 2020, Graduate program in Computer Science, KAIST)

Jiwon Jeon (B.S. 2020)

References

Available upon request

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